American Computer Science League

2025 Finals ● Short Problems ● Senior Division

1. Boolean Algebra

Which ordered triple makes the following Boolean expression FALSE?

$$\overline{A}(\overline{B} + C) + \overline{AB}(C + \overline{A}) + \overline{ABC}$$

A. (0,0,0)
B. (0,1,0)
C. (1,1,1)
D. (0,0,1)
E. None of the above

2. Boolean Algebra

The following Boolean expression can be implemented as a Digital Electronic circuit with a single 2-input gate.

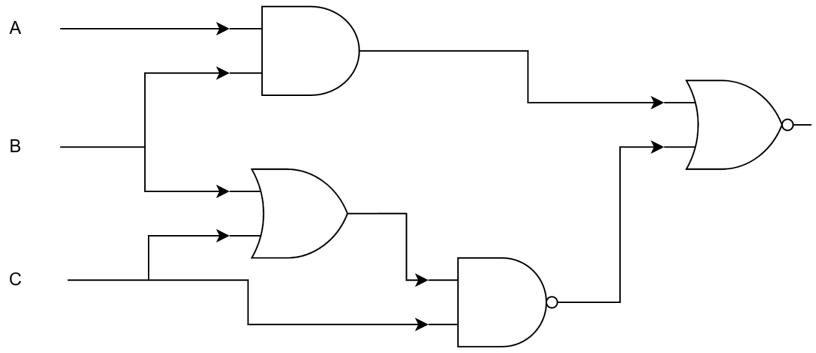
$$\overline{A + B}\,(\overline{A} + \overline{BC}) + \overline{A}\,\overline{B}(\overline{\overline{AB}C})$$

What gate is needed?

A. OR
B. AND
C. NOR
D. NAND
E. None of the above

3. Digital Electronics

How many ordered triples make the following digital circuit TRUE?



A. 2
B. 3
C. 4
D. 5
E. None of the above

4. Bit-String Flicking

Evaluate the following expression:

(LSHIFT-1 (RCIRC-2 (RSHIFT-3 (LCIRC-1 (NOT 10110)))))

A. 00000
B. 10000
C. 01001
D. 10010
E. None of the above

5. Bit-String Flicking

Solve for X, a bit-string of 5 bits.

(LCIRC-2 X) OR (RSHIFT-1 (RCIRC-2 (X AND 01110))) = (LSHIFT-2 (11010 OR X))

A. 01**0
B. 1*001
C. 0010*
D. 00*1*
E. None of the above

6. Recursive Functions

Find $f(10, 10)$, given:

$$f(x,y) = \begin{cases} 2 \cdot f(x-3,\ y+1) - 3 & \text{if } xy \geq 50 \\ f(x+1,\ y-4) + 1 & \text{if } 25 < xy < 50 \\ x+y & \text{if } xy \leq 25 \end{cases}$$

A. 6
B. 10
C. 18
D. 39
E. None of the above

7. Recursive Functions

Find $f(f(f(50)))$, given:

$$f(x) = \begin{cases} f\left(\frac{x}{2} - 1\right) + 3 & \text{if } x \text{ is even} \\ f(x-4) - 1 & \text{if } x \text{ is odd and } x \geq 8 \\ \left[\frac{x}{3}\right] & \text{otherwise} \end{cases}$$

Note: $[a]$ is the greatest integer less than or equal to a

A. -1
B. 0
C. 2
D. 5
E. None of the above

8. Prefix/Infix/Postfix Notation

Define two unary operators:

- ! as factorial (e.g. !3 = 3*2*1)
- % as the greatest integer of its square root (e.g. %7 = 2)

Evaluate the following prefix expression if all numbers are single digits.

/ * ! * 2 5 % * 9 + 9 2 * ! 8 % * ^ 2 3 5

A. 15
B. 30
C. 135
D. 405
E. None of the above

9. Prefix/Infix/Postfix Notation

Convert the following postfix expression to prefix if all numbers are single digits.

2 3 + 5 2 5 3 * + * − 8 4 / −

A. − − + 2 3 * 5 + 2 * 5 3 / 8 4
B. − − 2 + 3 * 5 + 2 * 5 3 / 8 4
C. − − + 2 3 5 * + 2 * 5 3 / 8 4
D. − − + 2 * 3 5 + * 2 5 3 / 8 4
E. None of the above

10. Computer Number Systems

What is the value of the following expression in hexadecimal?

$100_2 * (475_8 + 567_8)$

A. 5320
B. 2B4
C. 568
D. AD0
E. None of the above

11. Computer Number Systems

At a banquet there were 62_8 meals served: 11110_2 chose salmon @ $\$12_{16}$ per meal and 14_{16} chose beef @ $\$30_8$ per meal. What was the total food bill excluding the tip in hexadecimal?

A. 3E8
B. 1020
C. 3FC
D. 1774
E. None of the above

12. Data Structures

Given a binary search tree with the preorder traversal of **MICAIUSN** and the inorder traversal of **ACIIMNSU**, what is the postorder traversal?

A. AICIMUNS
B. ACIINSUM
C. AICIUNSM
D. AICINSUM
E. None of the above

13. Data Structures

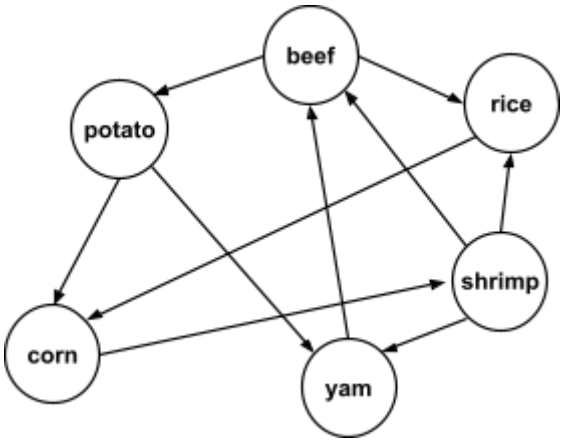
Follow the algorithm below on an initially empty stack. Which string will result in a single item of a "#" on the stack, without at any point trying to pop from an empty stack?

```
for each character in the string
     if the character is a letter
          push(character)
     else
          x = pop()
          y = pop()
          push("#")
     end if
next character
```

A. a * b c d e / + / f −
B. a b + c d / e / f −
C. a b c d e f − + * / ^
D. a b * c d / e f / * + ^
E. None of the above

14. Graph Theory

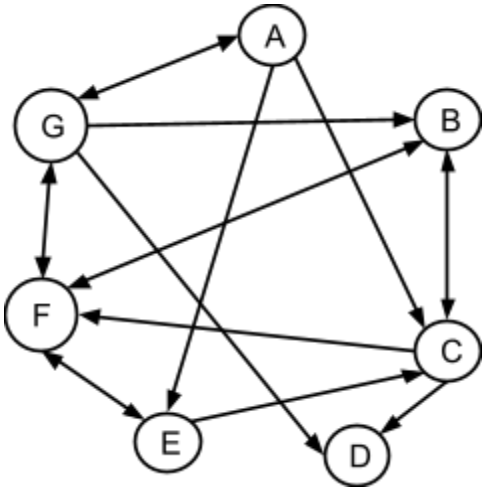
The chef wants to make meals with only 3 ingredients tonight with the items connected below. All three must form a path. How many different meals can be prepared?



A. 9
B. 11
C. 13
D. 15
E. None of the above

15. Graph Theory

How many cycles are there from F that do not contain A or C in the directed graph below?



A. 3
B. 4
C. 5
D. 6
E. None of the above

16. What Does This Program Do?

What is the output when the following program is executed?

```
num = 2025
x = 1
while num != 0
      x = x + 1
      if num % x == 0 then
            n = num
            t = 0
            while n % x == 0
                  t = t + 1
                  n = n / x
            end while
            output x,t
            num = num / x ^ t
      end if
end while
```

A. 3 4
   5 2
B. 5 2
   3 4
C. 2 5
   4 3
D. 2 1
   3 4
   4 1
   5 2
E. None of the above

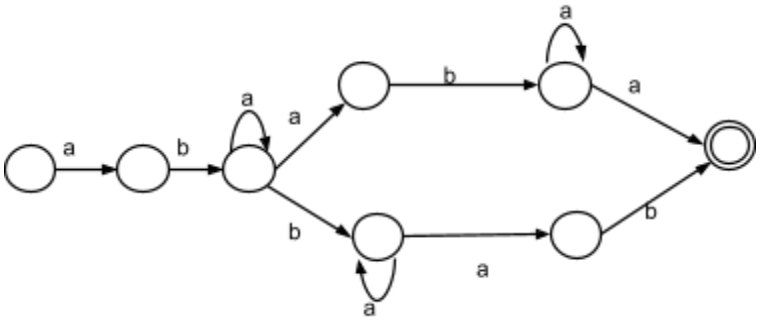
17. LISP

Given that x = '(a (b c) (d (e f))), which of the following expressions will evaluate to ((e f))?

A. (CDDDR x)
B. (CDDADR x)
C. (CDADDR x)
D. (CADDR x)
E. None of the above

18. FSAs and Regular Expressions

What is a regular expression that represents the FSA below?



A. aba* (aba* ∪ ba*ab)
B. aba* (ba*a ∪ ba*ab)
C. aba* (aba*a ∪ ba*ab)
D. aba (a*aba*a ∪ ba*ab)
E. None of the above

19. FSAs and Regular Expressions

Which of the following strings are accepted by this regular expression?

[abc][aeiou]*[lx]*[^g]*[o-x]*

a) afghan b) bulldog c) collie
d) boxer e) corgi f) akita
g) husky h) poodle

A. a, d, e
B. b, c, g
C. c, d, f
D. d, e, h
E. None of the above

20. Assembly Language Programming

What is the output when the following program is executed?

```
T       DC      0
N       DC      2025
        LOAD    N
        STORE   NUM
TOP     DIV     =10
        STORE   Q
        MULT    =10
        SUB     N
        MULT    =-1
        ADD     T
        STORE   T
        LOAD    Q
        BE      STOP
        STORE   N
        BU      TOP
STOP    LOAD    NUM
        DIV     T
        STORE   ANS
        PRINT   ANS
        END
```

A. 2025
B. 225
C. 200
D. 25
E. None of the above